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Department of Mathematics

LECTURE NOTES

Advanced Calculus and Numerical Methods (21MAT21)

Module-4 (Numerical Methods-1)

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Numerical solution of transcendental Equation:-

Definition:- An equation which contains algebraic, trigonometric, Exponential, logarithmic, etc, which is called the transcendental equation.

Ex:- $3x^2 - 7 \log x + 2e^{-x} + \sin x - 1 = 0$

To get the solution for the transcendental equation we have the following methods.

1. Regular - falsi method or false position method.
2. Newton - Rapson method.

1. Regular - falsi method:-

Working rule:-

1. Write the given transcendental equation as $f(x)=0$ and identify $f(x)$.
2. Choose x_0 & x_1 for which $f(x_0) < 0$, $f(x_1) > 0$.

3. Let x_2 be the required real root,

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

$$\text{if } f(x_2) < 0$$

then the root lies between

$$x_3 = \frac{x_2 f(x_1) - x_1 f(x_2)}{f(x_1) - f(x_2)}$$

→ Given if $f(x_3) < 0$

$$\text{then } x_4 = \frac{x_3 f(x_1) - x_1 f(x_3)}{f(x_1) - f(x_3)}$$

Continue this process until it reach the real roots.

1. Find the root of $x^3 - 9x + 1 = 0$ by Regular falsi method by four iteration.

Sol:- Given, $x^3 - 9x + 1 = 0$

$$\Rightarrow f(x) = 0$$

$$\therefore f(x) = x^3 - 9x + 1$$

and given $x_0 = 2, x_1 = 3$

$$f(x_0) = 2^3 - 9(2) + 1 = -9 < 0$$

$$f(x_1) = 3^3 - 9(3) + 1 = 1 > 0$$

$$\therefore x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

$$\therefore x_2 = \frac{(2)(1) - 3(-9)}{1+9} = \frac{2+27}{10}$$

$$\Rightarrow x_2 = \frac{29}{10}$$

$$\Rightarrow x_2 = 2.9$$

$$\therefore f(x_2) = (2.9)^3 - 9(2.9) + 1$$

$$f(x_2) = -0.711 < 0$$

$$\therefore x_3 = \frac{x_2 f(x_1) - x_1 f(x_2)}{f(x_1) - f(x_2)}$$

$$\Rightarrow x_3 = \frac{(2.9)(1) - 3(-0.711)}{1 + 0.711}$$

$$\Rightarrow x_3 = \frac{2.9 + 2.133}{1.711}$$

$$\Rightarrow x_3 = 2.9415$$

$$\therefore f(x_3) = 0.0022 < 0$$

$$x_4 = \frac{x_3 f(x_1) - x_1 f(x_3)}{f(x_1) - f(x_3)}$$

$$\Rightarrow x_4 = \frac{(2.9415)(1) - 3(-0.0022)}{1 + 0.0022}$$

$$\Rightarrow x_4 = \frac{3.0075}{1.0022}$$

$$\Rightarrow x_4 = 2.9997$$

$$\therefore f(x_4) = 0.002 < 0$$

$$\therefore x_5 = \frac{x_4 f(x_1) - x_1 f(x_4)}{f(x_1) - f(x_4)}$$

$$\Rightarrow x_5 = \frac{(2.9487)(1) - 3(-0.003)}{1 + 6.003}$$

$$\Rightarrow x_5 = \frac{2.9487}{1.003}$$

$$\Rightarrow x_5 = 2.9488$$

$\therefore$ The real root is 2.9488.

2. Find the root of equation: $xe^x = \cos x$ which lies in the interval $(0, 1)$ by regular falsi method for four decimal.

Sol: Given,

$$xe^x = \cos x$$

$$xe^x - \cos x = 0$$

$$f(x) = 0$$

$$\therefore f(x) = xe^x - \cos x$$

and given $x_0 = 0$, $x_1 = 1$

$$\therefore f(x_0) = -1 < 0$$

$$f(x_1) = 2.178 > 0$$

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

$$\Rightarrow x_2 = 0.3146$$

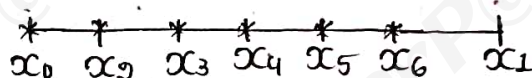
$$f(x_2) = -0.53 < 0$$

$$\therefore x_3 = \frac{x_2 f(x_1) - x_1 f(x_2)}{f(x_1) - f(x_2)}$$

$$\Rightarrow x_3 = 0.4685$$

$$f(x_3) = -0.1437 < 0$$

$$x_4 = \frac{x_3 f(x_1) - x_1 f(x_3)}{f(x_1) - f(x_3)}$$



$$\Rightarrow x_4 = 0.5011$$

$$\therefore f(x_4) = -0.0491 < 0$$

$$x_5 = \frac{x_4 f(x_1) - x_1 f(x_4)}{f(x_1) - f(x_4)}$$

$$\Rightarrow x_5 = 0.5124$$

$$\therefore f(x_5) = 0.0162 > 0$$

$$x_6 = \frac{x_5 f(x_1) - x_1 f(x_5)}{f(x_1) - f(x_5)}$$

$$\Rightarrow x_6 = 0.5160$$

$$\therefore f(x_6) = -0.0053 < 0$$

The real root is 0.5160

3. Find the real root of $xe^x = 3$ in the interval (1, 2) by using false method. find for 4 decimals.

Sol: Given,

$$xe^x = 3$$

$$xe^x - 3 = 0$$

$$f(x) = xe^x - 3 \text{ and given } x_0 = 1, x_1 = 2$$

$$\therefore f(x_0) = -0.28 < 0$$

$$f(x_1) = 11.7781 > 0$$

$$\therefore x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

$$\Rightarrow x_2 = 1.0232$$

$$\therefore f(x_2) = -0.1539 < 0$$

$$x_3 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

$$\Rightarrow x_3 = 1.0358$$

$$\therefore f(x_3) = -0.0818 < 0$$

$$\therefore x_4 = \frac{x_3 f(x_1) - x_1 f(x_3)}{f(x_1) - f(x_3)}$$

$$\Rightarrow x_4 = 1.0434$$

$$\therefore f(x_4) = -0.0437 < 0$$

$$x_5 = \frac{x_4 f(x_1) - x_1 f(x_4)}{f(x_1) - f(x_4)}$$

$$\Rightarrow x_5 = 1.0459$$

$$\therefore f(x_5) = -0.02341 < 0$$

$$\therefore x_6 = \frac{x_5 f(x_1) - x_1 f(x_5)}{f(x_1) - f(x_5)}$$

$$x_6 = 1.0478.$$

4. Using regular falsi method, find real root of $x^3 - 3x - 5 = 0$ in interval $(2, 3)$.

Sol: Given, $x^3 - 3x - 5 = 0$

$$\therefore f(x) = x^3 - 3x - 5$$

and given $x_0 = 2, x_1 = 3$

$$\therefore f(x_0) = (2)^3 - 3 \times 2 - 5 = -1 < 0$$

$$f(x_1) = 3^3 - 3(3) - 5 = 16 > 0$$

$$\therefore x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

$$\Rightarrow x_2 = 2.0588$$

$$\therefore f(x_2) = -0.3911 < 0$$

$$\therefore x_3 = \frac{x_2 f(x_1) - x_1 f(x_2)}{f(x_1) - f(x_2)}$$

$$\Rightarrow x_3 = 2.0813$$

$$\therefore f(x_3) = -0.1468 < 0$$

$$\therefore x_4 = \frac{x_3 f(x_1) - x_1 f(x_3)}{f(x_1) - f(x_3)}$$

$$\Rightarrow x_4 = 0.851$$

$$\therefore f(x_4) = -0.007 < 0$$

$$\therefore x_5 = \frac{x_4 f(x_1) - x_1 f(x_4)}{f(x_1) - f(x_4)}$$

$$\Rightarrow x_5 = 0.852$$

$$\therefore f(x_5) = -0.003$$

5. Find the real root of $xe^x - 2 = 0$ in the interval $(0, 1)$ by regular falsi method correct to three decimal places.

Sol:- Given, $xe^x - 2 = 0$

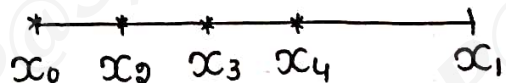
$$\Rightarrow f(x) = xe^x - 2 \text{ and given } x_0 = 0, x_1 = 1$$

$$\therefore f(x_0) = -2 < 0$$

$$f(x_1) = 0.718 > 0$$

$$\therefore x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

$$\Rightarrow x_2 = 0.736$$



$$\therefore f(x_2) = -0.464 < 0$$

$$\therefore x_3 = \frac{x_2 f(x_1) - x_1 f(x_2)}{f(x_1) - f(x_2)}$$

$$\Rightarrow x_3 = 0.840$$

$$\therefore f(x_3) = -0.054 < 0$$

$$\therefore x_4 = \frac{x_3 f(x_1) - x_1 f(x_3)}{f(x_1) - f(x_3)}$$

$$= \frac{(0.840 \times 0.718) + (1 \times 0.054)}{(0.718 + 0.054)}$$

$$x_4 = 0.851$$

$$\therefore f(x_4) = (0.851 e^{0.851}) - 2 = -0.007 < 0$$

$$\therefore x_5 = \frac{x_4 f(x_1) - x_1 f(x_4)}{f(x_1) - f(x_4)} = \frac{(0.851 \times 0.718) + (1 \times 0.007)}{0.718 + 0.007} = \boxed{0.852}$$

$$\therefore f(x_5) = (0.852 e^{0.852}) - 2 = -0.003$$

Newton - Raphson Method:-

1. Write the given transcendental equation as $f(x) = 0$.
2. Identify $f(x)$ and can find $f'(x)$.
3. Choose the nearest real root x_0 for which $f(x_0) < 0$.
4. Find the real root of the equation by the following way,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad \forall \quad n = 0, 1, 2, 3, \dots$$

i.e.,
$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

Continue this process until you reach an exact real root.

1. Find the real root of the equation $x \sin x + \cos x = 0$ near to the value $x = \pi$ using Newton's - Raphson method correct to four decimal.

Sol:- Given,

$$x \sin x + \cos x = 0$$

$$\Rightarrow f(x) = 0$$

$$\Rightarrow f(x) = x \sin x + \cos x$$

$$\Rightarrow f'(x) = x \cos x + \sin x - \sin x$$

$$\Rightarrow f'(x) = x \cos x$$

and given $x_0 = \pi$

$$\therefore x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$\Rightarrow x_1 = \pi - \left(\frac{-1}{-\pi} \right)$$

$$\Rightarrow x_1 = \pi - \frac{1}{\pi}$$

$$\Rightarrow x_1 = 3.14159 - \frac{1}{3.14159}$$

$$\Rightarrow x_1 = 3.14159 - 0.31831$$

$$\Rightarrow x_1 = 2.82328$$

$$\therefore x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$\Rightarrow x_2 = 2.82328 - \frac{(-0.0699)}{(-2.6840)}$$

$$\Rightarrow x_2 = 2.82328 - 0.0260$$

$$\Rightarrow x_2 = 2.79728$$

$$\therefore x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$\Rightarrow x_3 = 2.79728 - \frac{(-0.00056)}{(-2.6355)}$$

$$\Rightarrow x_3 = 2.79728 - 0.000212$$

$$\Rightarrow x_3 = 2.79707$$

$$\therefore x_4 = x_3 - \frac{f(x_3)}{f'(x_3)}$$

$$\Rightarrow x_4 = 2.79707 - \frac{0}{(-2.6358)}$$

$$\Rightarrow x_4 = 2.79707$$

2. Using the Newton - Raphson method, Find the real root of $3x = \cos x + 1$ correct to four decimal places.

Sol: Given,

$$3x = \cos x + 1$$

$$\cos x + 1 - 3x = 0$$

$$f(x) = \cos x + 1 - 3x$$

$$f'(x) = -\sin x - 3$$

$$\text{Let } x_0 = 1$$

$$\therefore x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \Rightarrow x_1 = 1 - \frac{(-1.45969)}{(-3.81414)} = 0.62$$

$$\therefore x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \Rightarrow x_2 = 0.62 - \frac{(-0.04612)}{(-3.5810)} = 0.6071$$

$$\therefore x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} \Rightarrow x_3 = 0.6071 - \frac{(0.0000)}{(-3.5704)} = 0.6071$$

$\therefore x = 0.6071$ is the real root.

3. Find the $\sqrt[3]{37}$ using Newton's - Raphson method.

Sol: Let

$$x = \sqrt[3]{37}$$

$$x^3 = 37$$

$$\Rightarrow x^3 - 37 = 0$$

$$f(x) = 0$$

$$f(x) = x^3 - 37$$

$$f'(x) = 3x^2$$

$$\text{Let } x_0 = 3$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 3 - \frac{(-10)}{27} = 3.3704$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 3.3704 - \frac{(1.2864)}{33.3207} = 3.3322$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 3.3327 - \frac{(0.0159)}{33.3207} = 3.3322$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = 3.3322 - \frac{(0.0000)}{33.3107} = 3.3322$$

$$\therefore x = \sqrt[3]{37} = 3.3322$$

4. Find root of $\tan x = x$, near to (4.5) using Newton-Raphson method.

Sol: Let $\tan x = x$

$$\Rightarrow x - \tan x = 0$$

$$\Rightarrow f(x) = 0$$

$$\therefore f(x) = x - \tan x$$

$$\Rightarrow f'(x) = 1 - \sec^2 x = 1 - 1 - \tan^2 x = -\tan^2 x$$

and also given $x_0 = 4.5$

$$\therefore x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$\Rightarrow x_1 = 4.5 - \frac{(-0.1373)}{-0.5043}$$

$$\Rightarrow x_1 = 4.4936$$

$$\therefore x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$\Rightarrow x_2 = 4.4936 - \frac{(-0.00039)}{-20.2271}$$

$$\Rightarrow x_2 = 4.4934$$

$$\therefore x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$\Rightarrow x_3 = 4.4934 - \frac{(0.0002)}{-20.1889}$$

$$\Rightarrow x_3 = 4.4934$$

$\therefore x = 4.4934$ is the real root.

5. Find the real root of $xe^x - \cos x$ correct to three decimal places by using Newton-Raphson method.

Sol:- $\Rightarrow f(x) = 0$

$$\therefore f(x) = xe^x - \cos x$$

$$\Rightarrow f'(x) = e^x + xe^x + \sin x$$

$$\text{Let } x_0 = 0.5$$

$$\therefore x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 0.5 - \frac{(-0.0532)}{2.905}$$

$$\Rightarrow x_1 = 0.518$$

$$\therefore x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 0.518 - \frac{(0.0007)}{3.0434}$$

$$\Rightarrow x_2 = 0.5178$$

$$\therefore x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 0.5178 - \frac{(0.00001)}{3.0423}$$

$$\Rightarrow x_3 = 0.5178$$

$\therefore x = 0.5178$ is the required real root.

6. Find the real root of the equation $x \log_{10} x = 1.2$ using Newton-Raphson method near to 2.5.

Sol: $\Rightarrow x \log_{10} x = 1.2$

$$\Rightarrow x \log_{10} x - 1.2 = 0$$

$$f(x) = 0$$

$$\Rightarrow f(x) = x \log_{10} x - 1.2$$

$$f(x) = \frac{x \log_e x}{\log_e 10} - 1.2$$

$$f(x) = \frac{x \log_e x}{2.3026} - 1.2$$

$$f(x) = (0.4343)x \log x - 1.2$$

$$\Rightarrow f'(x) = 0.4343 \left[\log x + \frac{1}{x} \right] - 0$$

$$f'(x) = 0.4343(1 + \log x)$$

$$\text{and } x_0 = 2.5$$

$$\therefore x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 2.5 - \frac{(-0.205)}{0.8322}$$

$$\Rightarrow x_1 = 2.7464$$

$$\therefore x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 2.7407$$

$$\therefore x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 2.7405$$

$$\therefore x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = 2.7406 - \left(\frac{0.0000}{0.8722} \right) = 2.7406$$

$\therefore x = 2.7407$ is the required real root.

Numerical Integration:-

1. Let $I = \int_a^b f(x) dx$, identify $a, b, n, y = f(x)$.

2. Find the step size $h = \frac{b-a}{n}$ and form the partition, from the given interval.

$$P = \{a = x_0, x_0 + h = x_1, x_1 + h = x_2, \dots, x_{n-1} + h = x_n = b\}$$

3. Find out the values of y are $y_0, y_1, y_2, \dots, y_n$ at $x_0, x_1, x_2, x_3, \dots, x_n$.

4. Use the following formula to find $\int_a^b f(x) dx$.

$\Rightarrow$ Simpson's $\frac{1}{3}$ rd Rule:-

$$\int_a^b f(x) dx = \frac{h}{3} \left[(y_0 + y_n) + 2(y_2 + y_4 + y_6 + \dots + y_{n-2}) + 4(y_1 + y_3 + y_5 + \dots + y_{n-1}) \right]$$

$\Rightarrow$ Simpson's $\frac{3}{8}$ th Rule:-

$$\int_a^b f(x) dx = \frac{3h}{8} \left[(y_0 + y_n) + 3(y_3 + y_6 + y_9 + \dots) + 2(y_1 + y_2 + y_4 + \dots) \right]$$

1. Evaluate $\int_0^1 \frac{1}{(1+x)} dx$ by taking six equal parts by using

i) Simpson's $\frac{1}{3}$ rd Rule ii) Simpson's $\frac{3}{8}$ th Rule.

Sol:- Let $I = \int_0^1 \frac{1}{(1+x)} dx$

$$\therefore a=0, b=1, n=6, \& y = \frac{1}{1+x}$$

$$\therefore h = \frac{b-a}{n} = \frac{1-0}{6} = \frac{1}{6}$$

$$\therefore P = \{a = x_0 = 0, x_1 = \frac{1}{6}, x_2 = \frac{2}{6}, x_3 = \frac{3}{6}, x_4 = \frac{4}{6}, x_5 = \frac{5}{6}, x_6 = 1\}$$

x	$1+x$	$y = \frac{1}{1+x}$
0	1	1 = y_0
$\frac{1}{6}$	$\frac{7}{6}$	0.8571 = y_1
$\frac{1}{3}$	$\frac{4}{3}$	0.7500 = y_2
$\frac{1}{2}$	$\frac{3}{2}$	0.6667 = y_3
$\frac{2}{3}$	$\frac{5}{3}$	0.6000 = y_4
$\frac{5}{6}$	$\frac{11}{6}$	0.5455 = y_5
1	2	0.5 = y_6

$$\begin{aligned}
 \Rightarrow \therefore \int_0^1 \frac{1}{1+x} dx &= \frac{h}{3} \left| (y_0 + y_6) + 2(y_2 + y_4) + 4(y_1 + y_3 + y_5) \right| \\
 &= \frac{\frac{1}{6}}{3} \left| (1 + 0.5) + 2(0.7500 + 0.6000) + 4(0.8571 + 0.6667 + 0.5455) \right| \\
 &= \frac{1}{18} \left| 1.5 + 2.7 + 8.2772 \right| \\
 &= \frac{12.4772}{18} \\
 \Rightarrow \int_0^1 \frac{1}{1+x} dx &= 0.6932 \\
 \Rightarrow \left| \log(1+x) \right|_0^1 &= 0.6932 \\
 \Rightarrow \log 2 - \log 1 &= 0.6932 \\
 \Rightarrow \log 2 &= 0.6932
 \end{aligned}$$

ii) Simpson's $\frac{3}{8}$ th Rule:-

$$\int_0^1 \frac{1}{1+x} dx = \frac{3h}{8} \left[(y_0 + y_6) + 2(y_3) + 3(y_1 + y_2 + y_4 + y_5) \right]$$

$$= \frac{3 \cdot \frac{1}{8}}{8} \left[(1+0.5) + 2(0.6667) + 3(0.8571 + 0.7500 + 0.6000 + 0.5455) \right]$$

$$= \frac{1}{16} \left[1.5 + 1.3334 + 8.2578 \right]$$

$$= 0.0625(11.0912) = 0.6932$$

$$\Rightarrow \int_0^1 \frac{1}{1+x} dx = 0.6932$$

Q. Evaluate $\int_0^1 \frac{1}{1+x^2} dx$ by taking 7 ordinates using

i) Simpson's $\frac{1}{3}$ rd rule, ii) Simpson's $\frac{3}{8}$ th rule.

sol:- Given, $I = \int_0^1 \frac{1}{1+x^2} dx$

$$\therefore a=0, b=1, n=6 \text{ and } y = \frac{1}{1+x^2}$$

$$\therefore h = \frac{b-a}{n} = \frac{1-0}{6} = \frac{1}{6}$$

$$\therefore P = \left\{ x_0=0, x_1=\frac{1}{6}, x_2=\frac{2}{6}, x_3=\frac{3}{6}, x_4=\frac{4}{6}, x_5=\frac{5}{6}, x_6=1 \right\}$$

x	$1+x^2$	$y = \frac{1}{1+x^2}$
0	1	1 = y_0
$\frac{1}{6}$	$\frac{37}{36}$	0.9730 = y_1
$\frac{1}{3}$	$\frac{10}{9}$	0.9 = y_2
$\frac{1}{2}$	$\frac{5}{4}$	0.8 = y_3

$\frac{2}{3}$	$\frac{13}{9}$	$0.6923 = y_4$
$\frac{5}{6}$	$\frac{61}{36}$	$0.5902 = y_5$
1	2	$0.5 = y_6$

1) Simpson's $\frac{1}{3}$ rd Rule:-

$$\text{WKT, } \int_0^1 \frac{1}{1+x^2} dx = \frac{h}{3} \left[(y_0 + y_6) + 2(y_2 + y_4) + 4(y_1 + y_3 + y_5) \right]$$

$$= \frac{\frac{1}{6}}{3} \left[(1 + 0.5) + 2(0.9 + 0.6923) + 4(0.9730 + 0.8 + 0.5902) \right]$$

$$= \frac{1}{18} [1.5 + 3.1846 + 9.4528]$$

$$= \frac{14.1374}{18} = 0.7854$$

$$\Rightarrow \int_0^1 \frac{1}{1+x^2} dx = 0.7854$$

$$\Rightarrow [\tan^{-1}x]_0^1 = 0.7854$$

$$\Rightarrow \tan^{-1}(1) - \tan^{-1}(0) = 0.7854$$

$$\Rightarrow \frac{\pi}{4} - 0 = 0.7854$$

$$\Rightarrow \pi = 4 \times 0.7854$$

$$\Rightarrow \pi = 3.1416$$

2) Simpson's $\frac{3}{8}$ th rule:-

$$\text{WKT, } \int_0^1 \frac{1}{1+x^2} dx = \frac{3h}{8} \left[(y_0 + y_6) + 2(y_3) + 3(y_1 + y_2 + y_4 + y_5) \right]$$

$$= \frac{\frac{2}{8^2}}{8} \left[(1+0.5) + 2(0.8) + 3(0.9730 + 0.9 + 0.6923 + 0.5902) \right]$$

$$= \frac{1}{16} [1.5 + 1.6 + 9.4665]$$

$$= \frac{12.5600}{16} = 0.7854$$

$$\int_0^1 \frac{1}{1+x^2} dx = 0.7854$$

3. Evaluate $\int_0^5 \frac{dx}{4x+5}$ using Simpson's $\frac{1}{3}$ rd rule dividing the interval into 10 equal parts.

Sol:- Given,

$$I = \int_0^5 \frac{dx}{4x+5}$$

where $a=0$, $b=5$, $n=10$ & $y = \frac{1}{4x+5}$

$$\therefore h = \frac{b-a}{n} = \frac{5-0}{10} = \frac{5}{10} = 0.5$$

$$P = \left\{ x_0=0, x_1=0.5, x_2=1, x_3=1.5, x_4=2, x_5=2.5, x_6=3, x_7=3.5, x_8=4, x_9=4.5, x_{10}=5 \right\}$$

x	$4x+5$	$y = \frac{1}{4x+5}$
0	5	$0.2 = y_0$
0.5	7	$0.1429 = y_1$
1	9	$0.1111 = y_2$
1.5	11	$0.0909 = y_3$
2.0	13	$0.0769 = y_4$
2.5	15	$0.0667 = y_5$
3	17	$0.0588 = y_6$
3.5	19	$0.0526 = y_7$

4	21	0.0476 = y ₈
4.5	23	0.0435 = y ₉
5	25	0.0400 = y ₁₀

∴ WKT,

$$\int_0^5 \frac{1}{4x+5} dx = \frac{h}{3} \left[(y_0 + y_{10}) + 2(y_2 + y_4 + y_6 + y_8) + 4(y_1 + y_3 + y_5 + y_7 + y_9) \right]$$

$$= \frac{0.5}{3} \left[(0.2 + 0.04) + 2(0.1111 + 0.0769 + 0.0588 + 0.0476) + 4(0.1429 + 0.0909 + 0.0669 + 0.0526 + 0.0435) \right]$$

$$= \frac{0.5}{3} [0.24 + 0.5888 + 1.5872] = 0.4027$$

$$\Rightarrow \int_0^5 \frac{1}{4x+5} dx = 0.4027.$$

4. Evaluate $\int_4^{5.2} \log x dx$ by using Simpson's $\frac{3}{8}$ th rule by taking 4 ordinates.

Sol:- Given, $I = \int_4^{5.2} \log x dx$

Where $a=4$, $b=5.2$, $n=6$ & $y = \log x$

$$\therefore h = \frac{b-a}{n} = \frac{5.2-4}{6} = \frac{1.2}{6} = 0.2$$

$$P = \left\{ x_0 = 4, x_1 = 4.2, x_2 = 4.4, x_3 = 4.6, x_4 = 4.8, x_5 = 5, x_6 = 5.2 \right\}$$

x	$y = \log_e x$
4	1.3863 = y_0
4.2	1.4351 = y_1
4.4	1.4816 = y_2
4.6	1.5261 = y_3
4.8	1.5686 = y_4
5	1.6094 = y_5
5.2	1.6481 = y_6

$$\begin{aligned}
 \therefore \int_4^{5.2} \log x \, dx &= \frac{3h}{8} \left[(y_0 + y_6) + 2(y_3) + 3(y_1 + y_2 + y_4 + y_5) \right] \\
 &= \frac{3 \times (0.2)}{8} \left[(1.3863 + 1.6481) + 2(1.5261) + 3(1.4351 + 1.4816 + 1.5686 + 1.6094) \right] \\
 &= \frac{0.6}{8} \left[3.0350 + 3.0522 + 18.284 \right] = 1.8278
 \end{aligned}$$

$$\Rightarrow \int_4^{5.2} \log x \, dx = 1.8278$$

5. Evaluate $\int_0^{\pi/2} \sqrt{\cos \theta} \, d\theta$ by using Simpson's $\frac{1}{3}$ rd rule by taking 7 ordinates.

sol:- Given, $I = \int_0^{\pi/2} \sqrt{\cos \theta} \, d\theta$

$$\therefore a = 0, b = \pi/2 = 90^\circ, n = 6, y = \sqrt{\cos \theta}$$

$$\therefore h = \frac{b-a}{n} = \frac{90-0}{6} = 15$$

$$h = \frac{b-a}{n} = \frac{\pi/2 - 0}{6} = \frac{\pi}{12} = \frac{3.1429}{12} = 0.2619$$

$$P = \left\{ \theta_0 = 0, \theta_1 = 15^\circ, \theta_2 = 30^\circ, \theta_3 = 45^\circ, \theta_4 = 60^\circ, \theta_5 = 75^\circ, \theta_6 = 90^\circ \right\}$$

θ	$y = \sqrt{\cos \theta}$	
0	1	$= y_0$
15	0.9828	$= y_1$
30	0.9306	$= y_2$
45	0.8409	$= y_3$
60	0.7071	$= y_4$
75	0.5087	$= y_5$
90	0	$= y_6$

$$\begin{aligned}
 \therefore \int_0^{\frac{\pi}{2}} \sqrt{\cos \theta} \cdot d\theta &= \frac{h}{3} \left[(y_0 + y_6) + 2(y_2 + y_4) + 4(y_1 + y_3 + y_5) \right] \\
 &= \frac{0.2619}{3} \left[(1 + 0) + 2(0.9306 + 0.7071) + 4(0.9828 + 0.8409 + 0.5087) \right] \\
 &= \frac{0.2619}{3} \left[1 + 3.2754 + 9.3296 \right] \\
 \Rightarrow \int_0^{\frac{\pi}{2}} \sqrt{\cos \theta} \, d\theta &= 1.1877
 \end{aligned}$$

6. Evaluate $\int_0^{\pi/2} \sqrt{\sin \theta} d\theta$ by using Simpson's $\frac{1}{3}$ rd rule by taking 7 ordinates.

Given, $I = \int_0^{\pi/2} \sqrt{\sin \theta} d\theta$

$\therefore a = 0, b = \frac{\pi}{2} = 90^\circ, n = 6, y = \sqrt{\sin \theta}, h = \frac{\pi/2 - 0}{6} = \frac{\pi}{12} = 15^\circ = 0.2619$

θ	$y = \sqrt{\sin \theta}$
0	0 = y_0
15	0.5087 = y_1
30	0.7071 = y_2
45	0.8409 = y_3
60	0.9306 = y_4
75	0.9828 = y_5
90	1 = y_6

$\therefore$ WKT,

$$\int_0^{\pi/2} \sqrt{\sin \theta} d\theta = \frac{h}{3} \left[(y_0 + y_6) + 2(y_2 + y_4) + 4(y_1 + y_3 + y_5) \right]$$

$$= \frac{0.2619}{3} \left[(0 + 1) + 2(0.7071 + 0.9306) + 4(0.5087 + 0.8409 + 0.9828) \right]$$

$$= \frac{0.2619}{3} \left[1 + 3.2754 + 9.3296 \right] = 1.1877$$

$$\Rightarrow \int_0^{\pi/2} \sqrt{\sin \theta} d\theta = 1.1877$$

Interpolation and divided differences:-

Suppose $y_0, y_1, y_2, y_3, \dots, y_n$ be the set of values of y at $x = x_0, x_1, x_2, \dots, x_n$ with equal step size 'h' then the evaluation of y between the given interval (x_0, x_n) is called an Interpolation and outside the given interval is called Extrapolation.

Forward and Backward differences:-

Let y_0, y_1, y_2, y_3, y_4 be the set of values of y at x_0, x_1, x_2, x_3, x_4 , then the forward difference table is defined as

forward difference table:-

x	y	Δ^1	Δ^2	Δ^3	Δ^4
x_0	y_0	$y_1 - y_0 = \Delta y_0$			
x_1	y_1		$\Delta y_1 - \Delta y_0 = \Delta^2 y_0$		
		$y_2 - y_1 = \Delta y_1$		$\Delta^2 y_1 - \Delta^2 y_0 = \Delta^3 y_0$	
x_2	y_2		$\Delta y_2 - \Delta y_1 = \Delta^2 y_1$		$\Delta^3 y_1 - \Delta^3 y_0 = \Delta^4 y_0$
		$y_3 - y_2 = \Delta y_2$		$\Delta^2 y_2 - \Delta^2 y_1 = \Delta^3 y_1$	
x_3	y_3		$\Delta y_3 - \Delta y_2 = \Delta^2 y_2$		
		$y_4 - y_3 = \Delta y_3$			
x_4	y_4				

where Δ (delta) is called forward difference operator and the $y_n, \nabla y_n, \nabla^2 y_n, \nabla^3 y_n, \nabla^4 y_n, \dots$ are called leading backward difference.

Newton's Forward interpolation formula:-

It is a formula to be used to find the value of y , then the value of x is near to x_0 .

$$y(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!} \Delta^4 y_0 + \dots$$
$$\Rightarrow p = \frac{x - x_0}{h}$$

Newton's Forward interpolation formula:-

It is a formula to be used to find the value of y , when x is near to x_n .

$$y(x) = y_n + p\nabla y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n + \frac{p(p+1)(p+2)(p+3)}{4!} \nabla^4 y_n + \dots$$

where, $p = \frac{x - x_h}{h}$

1. Using Newton's forward interpolation, find y at $x=5$, from the data.

x	4	6	8	10
y	1	3	8	16

Sol:

x	y	Δ	Δ^2	Δ^3
x_0	$y_0 = 1$			
x_1	$y_1 = 3$	$2 = \Delta y_0$		
x_2	$y_2 = 8$	5	$3 = \Delta^2 y_0$	
x_3	$y_3 = 16$	8	3	$0 = \Delta^3 y_0$

To find $y(5)$, since 5 is near to $x_0 = 4$

$$\therefore p = \frac{x - x_0}{h}$$

$$\Rightarrow p = \frac{5 - 4}{2} = \frac{1}{2}$$

$$\Rightarrow p = 0.5$$

$\therefore$ By the Newton forward Interpolation formula.

WKT,

$$y(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0$$

$$\Rightarrow y(5) = 1 + (0.5)(2) + \frac{(0.5)(0.5-1)}{2} (3) + \frac{(0.5)(0.5-1)(0.5-2)}{3 \times 2} (0)$$

$$\Rightarrow y(5) = 1 + 1 + \frac{(0.5)(-0.5)(3)}{2} + 0$$

$$\Rightarrow y(5) = 2 - 0.3750$$

$$\Rightarrow y(5) = 1.6250.$$

Q. Using Newtons backward interpolation formula, find the value of y at $x=6$, for given data.

x	1	2	3	4	5
y	1	-1	1	-1	1

Sol:-

x	y	∇	∇^2	∇^3	∇^4
$x_0 = 1$	$y_0 = 1$				
$x_1 = 2$	$y_1 = -1$	-2			
$x_2 = 3$	$y_2 = 1$	2	4		
$x_3 = 4$	$y_3 = -1$	-2	-4	-8	
$x_4 = 5$	$y_4 = 1$	2	4	8	16

$4 = \nabla^2 y_4$ $8 = \nabla^3 y_4$ $16 = \nabla^4 y_4$

To find $y(6)$:

Since 6 is near to $x_4 = 5$

$$\therefore p = \frac{x - x_4}{h} \Rightarrow p = \frac{6 - 5}{1} = 1$$

$\therefore$ WKT

$$y(x) = y_4 + p \nabla y_4 + \frac{p(p+1)}{2!} \nabla^2 y_4 + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_4 + \frac{p(p+1)(p+2)(p+3)}{4!} \nabla^4 y_4$$

$$\Rightarrow y(6) = 1 + (1)(2) + \frac{(1)(2)(4)}{2!} + \frac{(1)(2)(3)}{3!} (8) + \frac{(1)(2)(3)(4)}{4!} (16)$$

$$\Rightarrow y(6) = 1 + 2 + 4 + 8 + 16$$

$$\Rightarrow \boxed{y(6) = 31}$$

3. The population of a town is given by a table. year 1955 to 1985

year	1951	1961	1971	1981	1991
population	19.6	39.65	58.81	72.21	94.61

Sol:-

x	y	I	II	III	IV
$x_0 = 1951$	$y_0 = 19.6$				
		$20.05 = \Delta y_0$			
1961	39.65		$-0.89 = \Delta^2 y_0$		
		19.16		$-4.87 = \Delta^3 y_0$	$= \Delta^4 y_0$
1971	58.81		-5.76		$19.63 = \nabla^4 y_4$
		13.40		$14.76 = \nabla^3 y_4$	
1981	72.21		$9 = \nabla^2 y_4$		
		22.40			
$x_4 = 1991$	$y_4 = 94.61$	$= \nabla y_4$			

To find $y(1955)$:-

Since 1955 is near to 1951

$$\therefore p = \frac{x - x_0}{h} = \frac{1955 - 1951}{10} = \frac{4}{10} = 0.4$$

∴ WKT,

$$y(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!} \Delta^4 y_0$$

$$y(1955) = 19.6 + (0.4)(20.05) + \frac{(0.4)(-0.6)(-0.89)}{2!} + \frac{(0.4)(-0.6)(-1.6)(-4.87)}{3!} + \frac{(0.4)(-0.6)(-1.6)(-2.6)(19.63)}{4!}$$

$$\Rightarrow y(1955) = 19.6 + 8.02 + 0.1068 - 0.3117 - 0.8166$$

$$\Rightarrow y(1955) = 26.6$$

To find $y(1985)$:-

Since 1985 is near to 1991

$$p = \frac{x - x_4}{h} = \frac{1985 - 1991}{10} = -0.6$$

∴ WKT,

$$y(x) = y_4 + p\nabla y_4 + \frac{p(p+1)}{2!} \nabla^2 y_4 + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_4 + \frac{p(p+1)(p+2)(p+3)}{4!} \nabla^4 y_4$$

$$y(1985) = 94.61 + (-0.6)(22.4) + \frac{(-0.6)(0.4)(9)}{2} + \frac{(-0.6)(0.4)(1.4)(14.76)}{6} + \frac{(-0.6)(0.4)(1.4)(2.4)(19.63)}{24}$$

$$y(1985) = 94.61 - 13.44 - 1.08 - 0.8265 - 0.6591$$

$$\Rightarrow y(1985) = 78.604$$

∴ The increased population from 1985 to 1955 is

$$= 78.604 - 26.6$$

$$= 52$$

4. The area of a circle corresponding to diameter D is given below.

D	80	85	90	95	100
A	5026	5674	6362	7088	7854

Find the area corresponding to the diameter 105 using appropriate interpolation.

Sol:-

x	y	I	II	III	IV
$x_0 = 80$	$5026 = y_0$				
85	5674	648	40		
90	6362	688	38	-2	
95	7088	726	$40 = \nabla^2 y_4$	$2 = \nabla^3 y_4$	$4 = \nabla^4 y_4$
$x_4 = 100$	$7854 = y_4$	$766 = \nabla y_4$			

To find $y(105)$:

Since 105 is near to 100

$$P = \frac{x - x_4}{h} = \frac{105 - 100}{5} = \frac{5}{5} = 1$$

$\therefore$ WKT,

$$y(x) = y_4 + P \nabla y_4 + \frac{P(P+1)}{2!} \nabla^2 y_4 + \frac{P(P+1)(P+2)}{3!} \nabla^3 y_4 +$$

$$\frac{P(P+1)(P+2)(P+3)}{4!} \nabla^4 y_4$$

$$y(105) = 7854 + (1) 766 + \frac{(1)(2)(40)}{2} + \frac{(1)(2)(3)(2)}{3 \times 2} + \frac{(1)(2)(3)(4)}{4 \times 3 \times 2} (4)$$

$$\Rightarrow y(105) = 7854 + 766 + 40 + 2 + 4$$

$$\Rightarrow y(105) = 8666$$

5. Using Newton's backward interpolation formula, find $y(12.5)$ for the following data.

x	10	11	12	13
y	22	24	28	34

Sol:

x	y	∇	∇^2	∇^3
$x_0 = 10$	22			
11	24	2		
12	28	4	2	
$x_3 = 13$	34 = y_3	6 = ∇y_3	2 = $\nabla^2 y_3$	0 = $\nabla^3 y_3$

To find $y(12.5)$

since 12.5 is near to 13

$$p = \frac{x - x_4}{h} = \frac{12.5 - 13}{1} = -0.5$$

$\therefore$ WKT,

$$y(x) = y_3 + p \nabla y_3 + \frac{p(p+1)}{2!} \nabla^2 y_3 + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_3$$

$$\Rightarrow y(12.5) = 34 + (-0.5)6 + \frac{(-0.5)(0.5)(2)}{2} + \frac{(-0.5)(0.5)(1.5)(0)}{6}$$

$$\Rightarrow y(12.5) = 34 - 3 - 0.25$$

$$\Rightarrow y(12.5) = 30.75$$

6. Using Newton's Interpolation formula, estimate the number of students who obtained marks 40 and 45 from the following table.

Marks	30-40	40-50	50-60	60-70	70-80
No. of students	31	42	51	35	31

Sol: Number of students scored less than 40 marks = 31

No. of students scored less than 50 marks = $31 + 42 = 73$

No. of students scored less than 60 marks = $31 + 42 + 51 = 124$

No. of students scored less than 70 marks = $31 + 42 + 51 + 35 = 159$

No. of students scored less than 80 marks = $31 + 42 + 51 + 35 + 31 = 190$

x	y	Δ	Δ^2	Δ^3	Δ^4
$x_0 = 40$	$31 = y_0$				
50	73	$42 = \Delta y_0$			
60	124	51	$9 = \Delta^2 y_0$		
70	159	35	-16	$-25 = \Delta^3 y_0$	
		31	-4	12	$37 = \Delta^4 y_0$
$x_4 = 80$	$190 = y_4$				

To find $y(45)$:-

Since 45 is near to 40.

$$\therefore P = \frac{x - x_0}{h} = \frac{45 - 40}{10} = \frac{5}{10} = 0.5$$

$\therefore$ WKT,

$$y(x) = y_0 + P \Delta y_0 + \frac{P(P-1)}{2!} \Delta^2 y_0 + \frac{P(P-1)(P-2)}{3!} \Delta^3 y_0 + \frac{P(P-1)(P-2)(P-3)}{4!} \Delta^4 y_0$$

$$y(45) = 31 + (0.5)(42) + \frac{(0.5)(-0.5)(9)}{2} + \frac{(0.5)(-0.5)(-1.5)(-25)}{3 \times 2} +$$

$$\frac{(0.5)(0.5)(-1.5)(-2.5)(37)}{24}$$

$$\Rightarrow y(45) = 31 + 21 - 1.125 - 1.5625 - 1.4453$$

$$\Rightarrow y(45) = 47.86 \approx 48$$

$\therefore$ The number of students who scored 40 & 45 is

$$\rightarrow y(40) - y(45) = 48 - 31 = 17$$

Newton's Divided Differences for unequal intervals:-

Let the ordinates of x are $x_0, x_1, x_2, x_3, \dots, x_n$, corresponding to values of $y = f(x)$ are $f(x_0), f(x_1), f(x_2), f(x_3), \dots, f(x_n)$, then Newton's divided difference follows as given below for $n = 4$

x	$y = f(x)$	<u>I D.D</u>	<u>II D.D</u>	<u>III D.D</u>
x_0	$y = f(x_0)$	$\frac{f(x_1) - f(x_0)}{x_1 - x_0} = f(x_0, x_1)$		
x_1	$y = f(x_1)$	$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = f(x_1, x_2)$	$\frac{f(x_1, x_2) - f(x_0, x_1)}{x_2 - x_0} = f(x_0, x_1, x_2)$	
x_2	$y = f(x_2)$	$\frac{f(x_3) - f(x_2)}{x_3 - x_2} = f(x_2, x_3)$	$\frac{f(x_2, x_3) - f(x_1, x_2)}{x_3 - x_1} = f(x_1, x_2, x_3)$	$\frac{f(x_1, x_2, x_3) - f(x_0, x_1, x_2)}{x_3 - x_0} = f(x_0, x_1, x_2, x_3)$
x_3	$y = f(x_3)$	$\frac{f(x_4) - f(x_3)}{x_4 - x_3} = f(x_3, x_4)$	$\frac{f(x_3, x_4) - f(x_2, x_3)}{x_4 - x_2} = f(x_2, x_3, x_4)$	$\frac{f(x_2, x_3, x_4) - f(x_1, x_2, x_3)}{x_4 - x_1} = f(x_1, x_2, x_3, x_4)$
x_4	$y = f(x_4)$			$\frac{f(x_1, x_2, x_3, x_4) - f(x_0, x_1, x_2, x_3)}{x_4 - x_0} = f(x_0, x_1, x_2, x_3, x_4)$

The Newton's divided difference interpolation formula for $n=4$ is defined as

$$f(x) = f(x_0) + (x-x_0)f(x_0, x_1) + (x-x_0)(x-x_1)f(x_0, x_1, x_2) + (x-x_0)(x-x_1)(x-x_2)f(x_0, x_1, x_2, x_3) + (x-x_0)(x-x_1)(x-x_2)(x-x_3)f(x_0, x_1, x_2, x_3, x_4).$$

1. Use Newton's divided difference to find $f(9)$ for the following data.

x	5	7	11	13	17
$y = f(x)$	150	392	1452	2366	5202

Sol:- We prepare divided difference table.

x	$y = f(x)$	1 st D.D
$x_0 = 5$	$f(x_0) = 150$	
$x_1 = 7$	$f(x_1) = 392$	$f(x_0, x_1) = \frac{392 - 150}{7 - 5} = 121$
$x_2 = 11$	$f(x_2) = 1452$	$f(x_1, x_2) = \frac{1452 - 392}{11 - 7} = 265$
$x_3 = 13$	$f(x_3) = 2366$	$f(x_2, x_3) = \frac{2366 - 1452}{13 - 11} = 457$
$x_4 = 17$	$f(x_4) = 5202$	$f(x_3, x_4) = \frac{5202 - 2366}{17 - 13} = 709$

2 nd D.D	3 rd D.D
$f(x_0, x_1, x_2) = \frac{265 - 121}{11 - 5} = 24$ $f(x_1, x_2, x_3) = \frac{457 - 265}{13 - 7} = 32$ $f(x_2, x_3, x_4) = \frac{709 - 457}{17 - 11} = 42$	$f(x_0, x_1, x_2, x_3) = \frac{32 - 24}{13 - 5} = 1$ $f(x_1, x_2, x_3, x_4) = \frac{42 - 32}{17 - 7} = 1$

Now,

$$y = f(x) = f(x_0) + (x - x_0)f(x_0, x_1) + (x - x_0)(x - x_1)f(x_0, x_1, x_2) + (x - x_0)(x - x_1)(x - x_2)f(x_0, x_1, x_2, x_3).$$

$$\Rightarrow f(9) = 150 + (9 - 5)(121) + (9 - 5)(9 - 7)(24) + (9 - 5)(9 - 7)(9 - 11)(1)$$

$$\Rightarrow \boxed{f(9) = 810}$$

2. Construct the interpolating polynomial using Newton's divided difference formula for the given data. find $f(11)$.

x	2	4	5	6	8	10
f(x)	10	96	196	350	868	1746

Sol:-

x	y = f(x)	1 st D.D
$x_0 = 2$	$f(x_0) = 10$	$f(x_0, x_1) = \frac{96 - 10}{4 - 2} = 43$
$x_1 = 4$	$f(x_1) = 96$	$f(x_1, x_2) = \frac{196 - 96}{5 - 4} = 100$
$x_2 = 5$	$f(x_2) = 196$	$f(x_2, x_3) = \frac{350 - 196}{6 - 5} = 154$
$x_3 = 6$	$f(x_3) = 350$	$f(x_3, x_4) = \frac{868 - 350}{8 - 6} = 259$
$x_4 = 8$	$f(x_4) = 868$	$f(x_4, x_5) = \frac{1746 - 868}{10 - 8} = 439$
$x_5 = 10$	$f(x_5) = 1746$	

2 nd D.D	3 rd D.D
$f(x_0, x_1, x_2) = \frac{100 - 43}{5 - 2} = 19$ $f(x_1, x_2, x_3) = \frac{154 - 100}{6 - 4} = 27$ $f(x_2, x_3, x_4) = \frac{259 - 154}{8 - 5} = 35$ $f(x_3, x_4, x_5) = \frac{439 - 259}{10 - 6} = 45$	$f(x_0, x_1, x_2, x_3) = \frac{27 - 19}{6 - 2} = 2$ $f(x_1, x_2, x_3, x_4) = \frac{35 - 27}{8 - 4} = 2$ $f(x_2, x_3, x_4, x_5) = \frac{45 - 35}{10 - 5} = 2$

Now,

$$Y = f(x) = f(x_0) + (x - x_0) f(x_0, x_1) + (x - x_0)(x - x_1) f(x_0, x_1, x_2) \\ + (x - x_0)(x - x_1)(x - x_2) f(x_0, x_1, x_2, x_3)$$

$$\Rightarrow Y = 10 + (x - 2)43 + (x - 2)(x - 4)19 + (x - 2)(x - 4)(x - 5)2$$

$$\Rightarrow Y = 10 + 43x - 86 + (x - 2)(x - 4)[19 + 2x - 10]$$

$$\Rightarrow Y = 10 + 43x - 86 + (x^2 - 6x + 8)[2x + 9]$$

$$\Rightarrow Y = 10 + 43x - 86 + 2x^3 - 12x^2 + 16x + 9x^2 - 54x + 72$$

$$\Rightarrow Y = 2x^3 - 3x^2 + 5x - 4$$

We put $x = 11$

$$Y = 2(11)^3 - 3(11)^2 + 5(11) - 4$$

$$Y = 2350$$

Lagrange's Interpolation Formula:-

Let $y_0, y_1, y_2, \dots, y_n$ be the set of values of $y = f(x)$ corresponding to the values of $x_0, x_1, x_2, \dots, x_n$ then Lagrange's interpolation formula can be defined for 'n' tuple of values as

$$y(x) = \frac{(x-x_1)(x-x_2)(x-x_3)\dots(x-x_n)}{(x_0-x_1)(x_0-x_2)\dots(x_0-x_n)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)\dots(x-x_n)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)\dots(x_1-x_n)} y_1 + \dots + \frac{(x-x_0)(x-x_1)\dots(x-x_{n-1})}{(x_n-x_0)(x_n-x_1)\dots(x_n-x_{n-1})} y_n.$$

1. Using Lagrange's interpolation formula, find y at $x=10$, the data given below.

x	5	6	9	11
y	12	13	14	16

Sol:- Given,

$$x_0 = 5 \quad y_0 = 12$$

$$x_1 = 6 \quad y_1 = 13$$

$$x_2 = 9 \quad y_2 = 14$$

$$x_3 = 11 \quad y_3 = 16$$

WKT,

$$y(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} y_1 +$$

$$+ \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} y_3$$

$$\Rightarrow y(10) = \frac{(10-6)(10-9)(10-11)}{(5-6)(5-9)(5-11)} (12) + \frac{(10-5)(10-9)(10-11)}{(6-5)(6-9)(6-11)} (13) +$$

$$\frac{(10-5)(10-6)(10-11)}{(9-5)(9-6)(9-11)} (14) + \frac{(10-5)(10-6)(10-9)}{(11-5)(11-6)(11-9)} (16).$$

$$\Rightarrow y(10) = \frac{(4)(1)(-1)(12)}{(-1)(-4)(-6)} + \frac{(5)(1)(-1)(13)}{(1)(-3)(-5)} + \frac{(5)(4)(-1)(14)}{(4)(3)(-2)} + \frac{(5)(4)(1)(16)}{(6)(5)(2)}$$

$$\Rightarrow y(10) = 2 - 4.33 + 11.6 + 5.33$$

$$\Rightarrow \boxed{y(10) = 14.6}$$

2. Find the polynomial by using Lagrange's formula from the following data.

x	0	1	2	5
$y = f(x)$	2	3	12	147

Sol:-

$$x_0 = 0 \quad y_0 = 2$$

$$x_1 = 1 \quad y_1 = 3$$

$$x_2 = 2 \quad y_2 = 12$$

$$x_3 = 5 \quad y_3 = 147$$

$$y = f(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} y_1 +$$

$$\frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} y_3$$

$$\Rightarrow y = \frac{(x-1)(x-2)(x-5)}{(-1)(-2)(-5)} (2) + \frac{x(x-2)(x-5)}{(1)(-1)(-4)} (3) + \frac{x(x-1)(x-5)}{(2)(1)(-3)} (12) +$$

$$\frac{x(x-1)(x-2)}{(5)(4)(3)} (147)$$

$$\Rightarrow y = f(x) = \frac{1}{5} (x-1)(x-2)(x-5) + \frac{3}{4} x(x-2)(x-5) - 2x(x-1)(x-5) +$$

$$\frac{49}{20} x(x-1)(x-2)$$

$$\Rightarrow y = \frac{1}{20} \left[-4(x-1)(x^2-7x+10) + 15x(x^2-7x+10) - 40x(x^2-6x+5) + \right.$$

$$\left. 49x(x^2-3x+2) \right]$$

$$\Rightarrow y = \frac{1}{20} \left[-4(x^3-8x^2+17x-10) + 15(x^3-7x^2+10x) - 40(x^3-6x^2+5x) + \right.$$

$$\left. 49(x^3-3x^2+2x) \right]$$

$$\Rightarrow y = \frac{1}{20} \left[-4x^3 - 32x^2 - 68x + 40 - 15x^3 - 105x^2 + 150x - 40x^3 + \right.$$

$$\left. 240x^2 - 200x + 49x^3 - 147x^2 + 98x \right]$$

$$\Rightarrow y(x) = \frac{1}{20} \left[20x^3 + 20x^2 - 20x + 40 \right]$$

3. Using Lagrange's interpolation formula fit a polynomial which passes through points $(-1,0)$, $(1,2)$, $(2,9)$ & $(3,8)$ and hence estimate the value of y when $x = 2.2$.

$$\therefore x_0 = -1 \quad y_0 = 0$$

$$x_1 = 1 \quad y_1 = 2$$

$$x_2 = 2 \quad y_2 = 9$$

$$x_3 = 3 \quad y_3 = 8$$

$\therefore$ WKT,

$$y(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} y_1 +$$

$$\frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} y_3$$

$$\Rightarrow y(x) = \frac{(x-1)(x-2)(x-3)}{(-2)(-3)(-4)} (0) + \frac{(x+1)(x-2)(x-3)}{(2)(-1)(-2)} (2) +$$

$$\frac{(x+1)(x-1)(x-3)}{(2)(1)(-1)} (9) + \frac{(x+1)(x-1)(x-2)}{(4)(2)(1)} (8)$$

$$\Rightarrow y(x) = \frac{1}{24} (x+1)(x^2 - 5x + 6) - 3(x+1)(x^2 - 4x + 3) + (x+1)(x^2 - 3x + 2)$$

$$\Rightarrow y(x) = \frac{1}{24} \left[x^3 - 5x^2 + 6x + x^2 - 5x + 6 - 6(x^3 - 4x^2 + 3x + x^2 - 4x + 3) + 2(x^3 - 3x^2 + 2x + x^2 - 3x + 2) \right]$$

$$\Rightarrow y(x) = \frac{1}{24} \left[x^3 - 4x^2 + x + 6 - 6x^3 + 24x^2 - 18x - 6x^2 + 24x - 18 + 2x^3 - 6x^2 + 4x + 2x^2 - 6x + 4 \right]$$

$$\Rightarrow y(x) = \frac{1}{24} \left[-3x^3 + 10x^2 + 5x - 8 \right]$$

$$\Rightarrow y(x) = -\frac{3}{24} x^3 + \frac{5}{24} x^2 + \frac{5}{24} x - 4$$

$$\Rightarrow y(2.2) = -\frac{3}{24} (2.2)^3 + \frac{5}{24} (2.2)^2 - \frac{5}{24} (2.2) - 4$$

$$\Rightarrow y(2.2) = 9.728$$